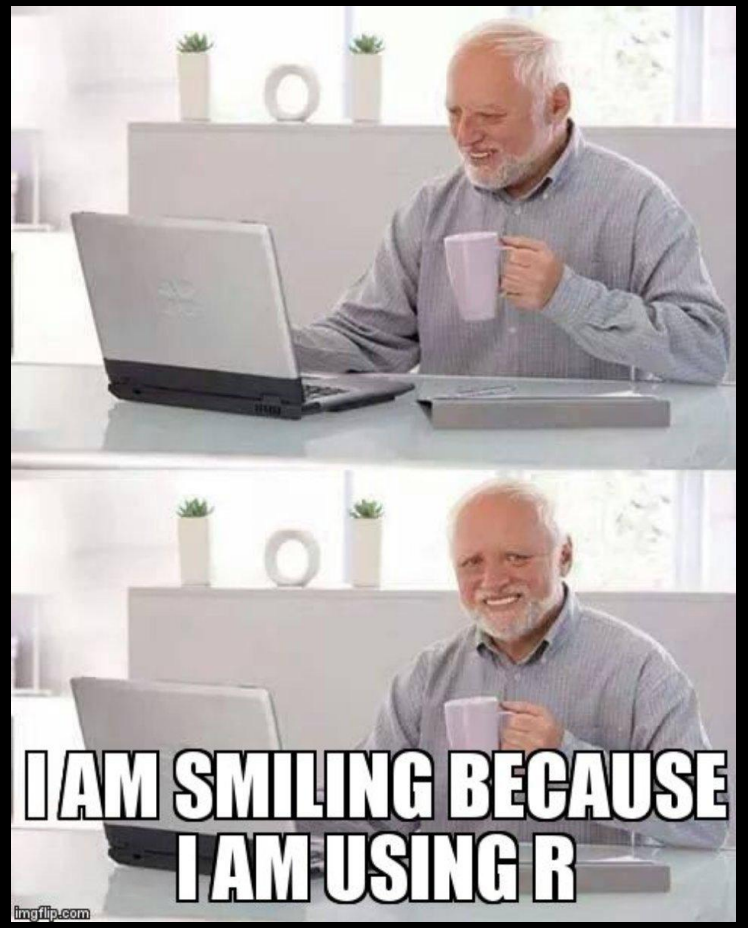


MES 503 Lab 11

Kayla Blincow, PhD

Announcements

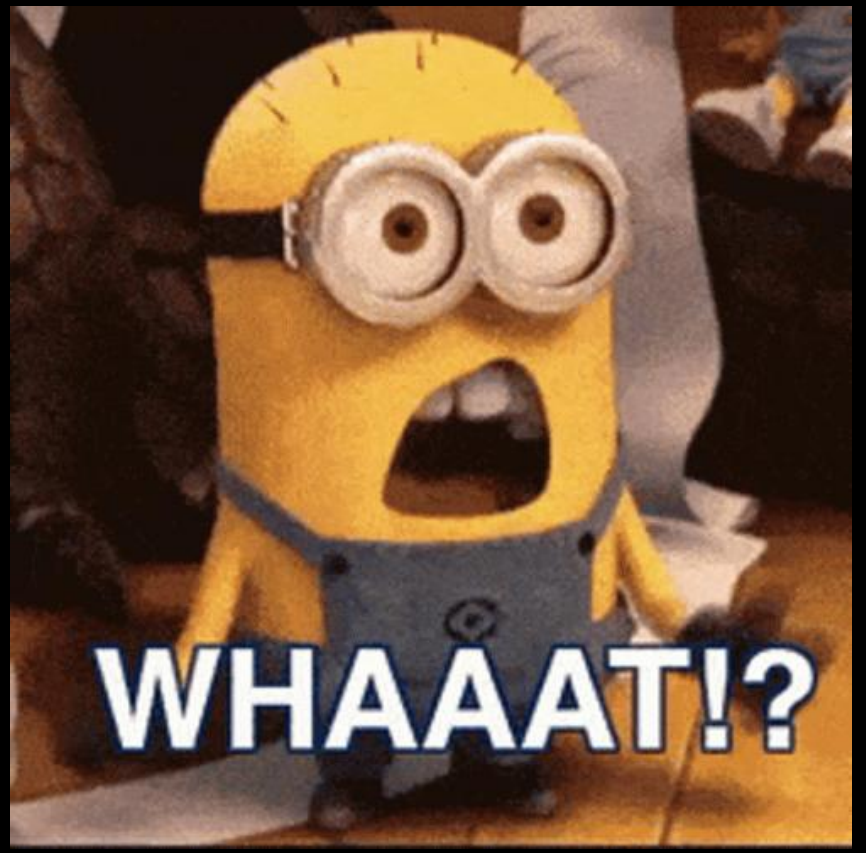
- No class on Monday!
- Quiz next Wednesday!
- Lab 4 due tomorrow!
- Next week bring your own dataset to work with

- Happy Halloweeeeeennn!

TENTATIVE SCHEDULE OF TOPICS – MES 503 Fall 2026

		Mondays (Lectures)		Wednesdays (Labs)	
Week	**Date**	**Lecture Topic**	**Date**	**Lab topic**	**Assessments**
1	8/17	Introduction to Statistics & Experimental Design I	8/19	Introduction to R and RStudio	
2	8/24	Introduction to Statistics & Experimental Design II	8/26	Intro to ggplot2	
3	8/31	Correlation	9/2	Correlation in R	Quiz 1
4	9/7	LABOR DAY (No Class)	9/9	Intro to Data wrangling	Lab 1 Due (9/10)
5	9/14	Statistical Hypothesis Testing & t tests	9/16	t Tests in R	
6	9/21	Linear Regression	9/23	Linear Regression in R	Quiz 2 (9/21) Lab 2 Due (9/24)
7	9/28	ANOVA	9/30	ANOVA in R	
8	10/5	ANOVA II (Two-Way & Nested) (virtual)	10/7	Two-Way & Nested ANOVA in R	Quiz 3
9	10/12	Multiple Linear Regression	10/14	Multiple Linear Regression in R	Lab 3 Due (10/15)
10	10/19	ANCOVA	10/21	ANCOVA in R	Quiz 4
11	10/26	Analysis of Frequencies	10/28	Analysis of Frequencies in R	Lab 4 Due (10/29)
12	11/2	Liberty Day – UVI Closed	11/4	Miscellaneous Fun R Topic	Quiz 5
13	11/9	Quantifying Variability/Figure and Data Sheet Design	11/11	Veterans Day – UVI Closed	
14	11/16	Reading for Stats Presentations	11/18	Reading for Stats Presentations	Lab 5 Due (11/17)
15	11/23	Applying this course to your thesis	11/25	Study Day	
16			12/3	**Final Exam**	

What type of test would you use with a **categorical explanatory variable** and a **categorical response variable**?

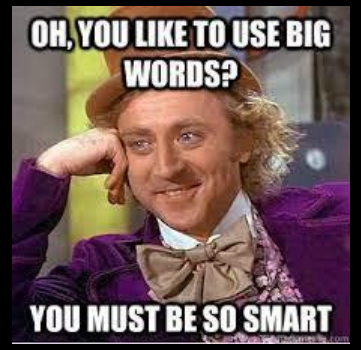


Enter Analysis of Frequencies!



What is a frequency?

- counts of observations that fall into categories





Types of Questions

- Does my distribution of frequencies across each category match the expected distribution across categories?
- Does the distribution of frequencies across one categorical variable affect the distribution of frequencies across another categorial variable?
- What is the likelihood that two distributions of frequencies come from the same population?

Research Scenario!

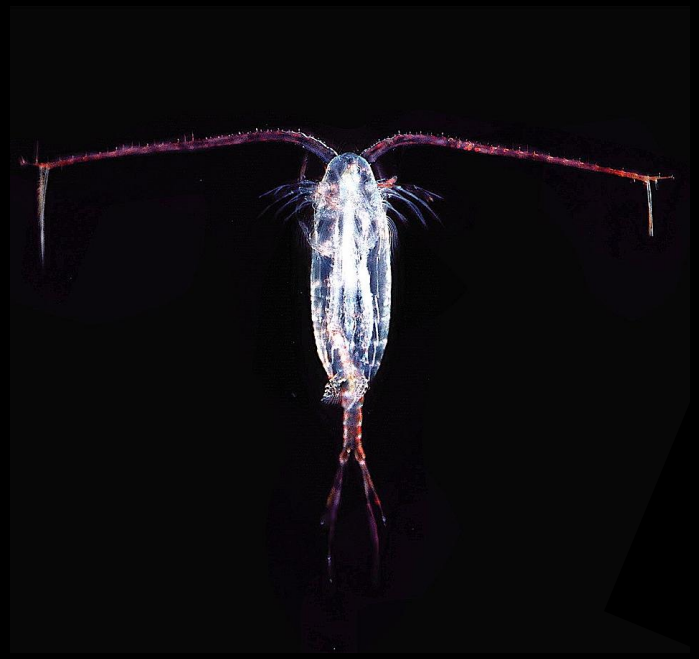
- We are doing some big ship science!
- Sample copepods during an El Nino year
- Count the number of each group





Meet the Copepods!

Calanoid



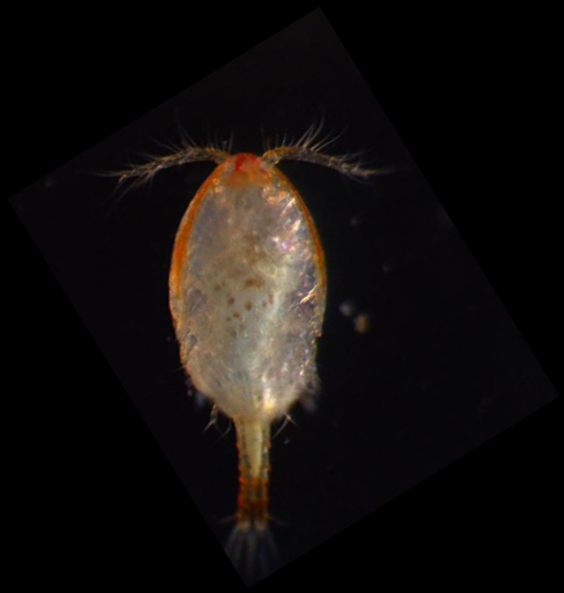
Harpacticoid



Cyclopoid



Poecilostomatoid



*note: apparently poecilotomes are now a suborder of cyclopoids, but we are ignoring that for the purposes of this lecture

Blackwater Manor
Coding Escape Room



https://kmblincow-the-blackwater-manor-incident.share.connect.posit.cloud/